Horizontal Tailplane External Nodal Forces

Load derivation from the V-n diagram to the ANSYS APDL load file — method, inputs and verification

Project	Parametric APDL-based horizontal tailplane structural analysis and optimisation
Aircraft	ATR 42-600 (flight loads) with the horizontal tailplane geometry of Palumbo et al. (2026)
Scope	Balancing horizontal-tail loads only, converted to external nodal forces for the FE model
Model	One half of the horizontal tailplane, Config A (14 ribs, 3 spars, 8 stringers)
Tools	MATLAB (steps 1 to 4), ANSYS Mechanical APDL (node export and load application)
Status	Preliminary. Values marked ASSUMED are not confirmed aircraft data.

1. Purpose and overview

This document records every step taken to produce the external nodal force files used by the ANSYS APDL horizontal tailplane (HTP) model. The chain starts from the aircraft flight envelope and ends with a set of `F,node,FZ,value` commands that reproduce the internal shear force and bending moment distribution of the tail. The approach follows the methodology outline of Palumbo et al. (2026): a polynomial load distribution whose coefficients are recovered by a least-norm least-squares solve.

Four MATLAB scripts run in sequence, each passing data to the next through a CSV file. ANSYS is used at two points only: once before step 4, to export the list of nodes that will carry the load, and once after step 4, to read the generated load file.

Step	What it produces	Script	Output file
1	Flight envelope and critical load factors	`vn_diagram.m`	`vn_load_cases.csv`
2	Balancing tail load L_H at every V-n point, both CG limits	`tail_load.m`	`tail_load_cases.csv`
3	Spanwise load, shear force and bending moment along the half-span	`htp_sfbm.m`	`htp_sfbm.csv`
–	List of FE nodes to be loaded (node number and span position)	APDL export block	`htp_nodes.csv`
4	External nodal forces and the APDL load file	`htp_nodal_forces.m`	`htp_loads_<case>.inp`

Sign convention used throughout: a positive force or load acts in the global **+Z** direction (upward). A negative tail load is therefore a download, and it produces a negative shear force and a negative bending moment at the root.

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2. Step 1 — V-n diagram and critical load factors

`vn_diagram.m` builds a preliminary manoeuvre and gust envelope for the ATR 42-600 at sea level, in equivalent airspeed, following the CS-25 / JAR-25 form. Its purpose is to identify the speed and load factor combinations at which the tail load is to be evaluated.

2.1 Input data

Quantity	Symbol	Value	Source
Maximum take-off weight	W	18 600 kg	ATR 42 parameter summary
Wing reference area	S	54.5 m^2	ATR 42 parameter summary

Quantity	Symbol	Value	Source
Wing span	b	24.57 m	ATR 42 parameter summary
Aspect ratio	AR	11.08	ATR 42 parameter summary
Mean aerodynamic chord	c_bar	2.285 m	ATR 42 parameter summary
Maximum operating speed	V_{MO}	250 kt	ATR 42 parameter summary
Maximum lift coefficient, clean	CL_{max}	1.5	ATR 42 parameter summary
Minimum lift coefficient, clean	$CL_{max,neg}$	-1.0	ASSUMED
Design cruise speed	V_C	= V_{MO} = 250 kt	ASSUMED
Design dive speed	V_D	= 1.25 V_C = 312.5 kt	CS-25.335 minimum margin
Airfoil efficiency factor	eta	0.95	ASSUMED

2.2 Method

The lift-curve slope is obtained from the Helmbold / DATCOM form for an unswept wing in incompressible flow:

```
a = 2 pi AR / ( 2 + sqrt( 4 + (AR/eta)^2 ) )    [1/rad]
```

The positive limit load factor comes from CS-25.337, with the weight in pounds and the result clamped between 2.5 and 3.8:

```
n_max = 2.1 + 24000 / (W_lb + 10000),   2.5 <= n_max <= 3.8
n_min = -1.0
```

The stall boundaries are the parabolas n = 0.5 rho_0 V^2 S CL / W, evaluated with CL_{max} and $CL_{max,neg}$. Corner speeds follow from their intersection with the load factor limits:

```
V_S1 = sqrt( 2W / (rho_0 S CL_max) )          and   V_A = V_S1 sqrt(n_max)
V_S,neg = sqrt( 2W / (rho_0 S |CL_max,neg|) ) and   V_G = V_S,neg sqrt(|n_min|)
```

Gust load factor increments use the Pratt alleviation factor, with the mass ratio mu:

```
mu = 2 (W/S) / (rho_0 c_bar g a)        and   K_g = 0.88 mu / (5.3 + mu)
delta_n = K_g rho_0 U_de V a S / (2W)
```

The design speed for maximum gust intensity V_B is taken from CS-25.335(d) and capped at V_C. The gust velocities used are 66, 50 and 25 ft/s EAS at V_B, V_C and V_D respectively, which is the JAR-25 change 15 form of the requirement.

> **Verify before submission.** The 66 / 50 / 25 ft/s gust velocities and the certification basis (JAR-25 change 15) should be checked against the applicable regulation for the ATR 42-600, because later amendments of FAR/CS-25.341 use a different gust formulation.

2.3 Results

V_{S1}	V_A	V_B	V_C	V_D	V_G	n_{max}	n_{min}
117 kt	188 kt	183 kt	250 kt	312.5 kt	144 kt	2.571	-1.00

The script writes twelve load cases to `vn_load_cases.csv`: the six manoeuvre envelope corners A, C, D, E, F and G, and the six gust intersections GustB+, GustC+, GustD+, GustB-, GustC- and GustD-. Each row carries the point name, speed in knots and in m/s EAS, load factor, weight and a description.

Discrepancy to report. The CS-25.337 formula gives n_{max} = 2.571 for an 18 600 kg aircraft, whereas the group parameter summary quotes 2.5. The scripts use 2.571; the override variable `n_max_override` at the top of the script forces 2.5 if the team prefers the quoted value.

3. Step 2 — Balancing horizontal-tail load

`tail_load.m` reads the twelve V-n points and computes, for each one, the horizontal-tail load required to hold the aircraft in pitch equilibrium. It does this at both the forward and the aft centre-of-gravity limit, giving 24 cases.

3.1 Derivation

For steady flight at small angles, neglecting thrust and drag couples, vertical force and moment equilibrium about the centre of gravity give:

```
L_w + L_H = n W                                   (vertical force)
L_w (x_cg - x_ac) c_bar + M_ac - L_H l_H = 0      (pitching moment, nose-up positive)
```

Eliminating the wing lift L_w yields the balancing tail load:

```
L_H = ( n W (x_cg - x_ac) c_bar + Cm_0 q S c_bar ) / ( l_H + (x_cg - x_ac) c_bar )
```

where q = 0.5 rho_0 V^2 is the sea-level dynamic pressure in equivalent airspeed, and M_ac = Cm_0 q S c_bar is the wing-body zero-lift pitching moment (nose-down, so Cm_0 is negative). The CG positions x_cg and the aerodynamic centre x_ac are fractions of the mean aerodynamic chord.

> **Note on the source equation.** The balancing tail load equation in the project reference omits the mean aerodynamic chord factor on the weight term. The form above is dimensionally consistent and reduces to the published equation once that factor is restored.

3.2 Input data

Quantity	Symbol	Value	Source
Tail arm, CG to tail aerodynamic centre	l_H	11.46 m	ATR 42 parameter summary
Forward CG limit	$x_{cg,fwd}$	0.168 c_bar (16.8 % MAC)	ATR 42 parameter summary
Aft CG limit	$x_{cg,aft}$	0.325 c_bar (32.5 % MAC)	ATR 42 parameter summary
Wing-body aerodynamic centre	x_{ac}	0.25 c_bar	Standard quarter-chord value
Wing-body zero-lift moment coefficient	Cm_0	-0.06	ASSUMED
Horizontal tail area (total, both halves)	S_H	19.69 m^2	Palumbo geometry; used only for the CL_H sanity check

The script also sweeps Cm_0 over -0.03, -0.06 and -0.09 to show how sensitive the result is to this assumed value, and it checks that the required tail lift coefficient stays well below unity.

3.3 Results and case selection

Forward CG produces a tail download, because the lift must act behind a CG that is ahead of the aerodynamic centre. Aft CG can produce an upload. Three cases were selected for the finite element analysis:

Case	V-n point	CG	V [kt EAS]	n [-]	L_H total [N]	Character
`D_fwd`	D — dive speed at n_{max}	forward	312.5	2.571	-18 286	Largest download — critical
`C_fwd`	C — cruise speed at n_{max}	forward	250.0	2.571	-14 509	Download at cruise
`A_aft`	A — stall meets n_{max}	aft	188.0	2.571	+3 224	Largest upload — load reversal

All 24 cases are written to `tail_load_cases.csv`, so any other point can be analysed by changing one line in step 3.

4. Step 3 — Spanwise load, shear force and bending moment

`htp_sfbm.m` distributes the half-tail load along the half-span and integrates it from tip to root to obtain the internal shear force and bending moment diagrams. These are the target distributions that the nodal forces must reproduce.

4.1 Geometry

Root chord	Tip chord	Half-span	Taper ratio	Dihedral	Ribs	Area per half
3.090 m	1.860 m	3.978 m	0.602	7 deg	14	9.85 m^2

The trailing edge is straight, so the whole taper is taken on the leading edge and the chord varies linearly from root to tip. Rib stations are equally spaced at 3978/13 = 306.0 mm.

4.2 Load distribution and integration

The load carried by one half is P = L_H/2. It is distributed using a Schrenk approximation, the average of a chord-proportional distribution and a quarter-ellipse, each normalised so that it integrates to unity over the half-span. The running load is then w(y) = P × shape(y), which carries exactly P.

```
shape_chord(y) = c(y) / integral of c over the half-span
shape_ell(y)   = (4 / (pi s)) sqrt( 1 - (y/s)^2 )
shape(y)       = 0.5 ( shape_chord + shape_ell )
```

Integrating inboard from the tip, with y measured from the root:

```
V(y) = P - integral from 0 to y of w     (shear force, V(0) = P, V(s) = 0)
M(y) = integral of V from y to s         (bending moment, M(s) = 0)
```

Because the loads are vertical in the global frame and the moment arms are horizontal, the 7 degree dihedral does not change V(y) or M(y). It affects the finite element geometry only. Torsion is not considered in this analysis, in line with the agreed project scope.

4.3 Results

The load centroid of the Schrenk distribution lies at 1.756 m from the root, which is 44.1 % of the half-span. The root bending moment is therefore P times that arm.

Case	Half load P [N]	Root shear V_{root} [N]	Root moment M_{root} [N m]
`D_fwd`	-9 143	-9 143	-16 058
`C_fwd`	-7 254	-7 254	-12 741
`A_aft`	+1 612	+1 612	+2 831

Built-in checks confirm V(0) = P, M(0) = P × 1.756 m and V = M = 0 at the tip. The script writes w, V and M at the 14 rib stations to `htp_sfbm.csv`.

5. Node export from ANSYS

Before the nodal forces can be computed, MATLAB needs to know which finite element nodes will carry the load and where they are along the span. This block is inserted into the APDL script **after** the final `NUMMRG, NODE` and `NUMCMP, NODE` commands and **before** `/SOLU`. Position matters: any renumbering or node merging performed after the export invalidates the node numbers.

```
! ---- Export loaded nodes for MATLAB (rib-station skin nodes) ----
CSYS,0                              ! global Cartesian
ALLSEL,ALL
*CFOPEN,htp_nodes,csv
*VWRITE
('node,y_mm,x_mm,z_mm')
*DO,I,2,N_RIB                       ! skip rib 1 (root, fixed)
    ALLSEL,ALL
    ESEL,S,MAT,,2                   ! skin shell elements only
    NSLE,S                          ! corner nodes of those elements
    NSEL,R,LOC,Y,Y_STA(I)-0.5,Y_STA(I)+0.5
    *GET,NN,NODE,0,COUNT
    NI = 0
    *DO,K,1,NN
        NI = NDNEXT(NI)
        *GET,YN,NODE,NI,LOC,Y
        *GET,XN,NODE,NI,LOC,X
        *GET,ZN,NODE,NI,LOC,Z
        *VWRITE,NI,YN,XN,ZN
        (F10.0,',',F12.4,',',F12.4,',',F12.4)
    *ENDDO
*ENDDO
*CFCLOS
ALLSEL,ALL
```

Three features of this block are deliberate

- **Material 2 selection.** `ESEL,S,MAT,,2` restricts the selection to the composite skin shell elements. Selecting nodes by area or by location alone also picks up the internal nodes of the BEAM188 stringer, spar-flange and rib-flange elements. Those nodes carry no load to the structure, and loading them loses that part of the applied force (see section 9).
- **Rib 1 is skipped.** The loop starts at I = 2 because rib 1 is the fixed root. A force applied there would pass straight into the support and would not load the structure.
- **Global coordinate system.** `CSYS,0` guarantees that the exported Y values are span positions in the global frame.

For the Config A model the export produces **1 052 nodes** across 13 loaded rib stations. The X and Z columns are written for checking only; MATLAB reads the first two columns.

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6. Step 4 — Conversion to external nodal forces

`htp_nodal_forces.m` is the core of the task. It converts the continuous shear force and bending moment distributions of step 3 into a discrete set of nodal forces that reproduces them.

6.1 Formulation

Following the methodology of Palumbo et al., the running load is written as a polynomial in normalised span position, here of degree three:

```
w(y) = sum over p of  a_p (y/s)^p,     p = 0 ... 3
```

Each rib station is assigned a tributary interval, bounded by the mid-points to its neighbours. The interval of the first loaded station extends inboard to the root, and that of the last station extends outboard to the tip, so the intervals cover the whole half-span without gaps. The total force at a station is the integral of w over its tributary interval, and it is shared equally between the nodes at that station:

```
F_station(g) = integral of w(y) dy over the tributary interval of station g
```

```
F_node       = F_station(g) / (number of nodes at station g)
```

The unknown coefficients a_p are then found by requiring that the nodal forces reproduce the target shear force and bending moment at a series of cuts, placed at the root and mid-way between each pair of stations:

```
V(cut) = sum of F_station over all stations outboard of the cut
M(cut) = sum of F_station (y_station - y_cut) over the same stations
```

This gives an over-determined system in four unknowns. The rows are scaled so that the shear force and bending moment equations are of comparable magnitude, the root cut is weighted by a factor of 100 so that the total load and the root moment are matched closely, and the system is solved with `lsqminnorm`, which returns the least-norm least-squares solution.

6.2 Built-in verification

The script recomputes the shear force and bending moment from the solved nodal forces and compares them with the targets. For case D_fwd:

Check	From nodal forces	Target	Agreement
Sum of all nodal forces	-9 142.8 N	-9 142.9 N	exact to rounding
Root bending moment	-16 057.5 N m	-16 057.5 N m	exact to rounding
Maximum shear error at a rib	879.6 N	—	9.6 % of root shear
Maximum moment error at a rib	134.9 N m	—	0.8 % of root moment

The shear force error at a rib station is expected and is not a defect. A rib carries a concentrated force, so the shear force steps across it, and the value computed at the rib itself differs from the smooth curve by up to half of that station force. The cuts between the ribs are matched exactly, and the total load and the root moment, which govern the overall equilibrium of the model, are matched to within rounding.

6.3 Output

- `htp_loads_<case>.inp` — one `F,node,FZ,value` command per loaded node, ready to be read by ANSYS.
- `htp_nodal_forces.csv` — the same data as node number, span position and force, for inspection.
- `htp_nodal_forces.png` — a three-panel check plot of the station forces and of the reconstructed shear force and bending moment against their targets.

The case is selected by the variable `case_to_export` at the top of the script. The node list is independent of the load case, so `htp_nodes.csv` is exported once and reused for all three cases.

7. Applying and verifying the loads in ANSYS

The generated load file is placed in the ANSYS working directory and read in the solution phase, after the root constraint and before SOLVE. The placeholder distributed pressure originally present in Block 9 of the APDL script is removed.

```
! ================================================================
!  BLOCK 9: BOUNDARY CONDITIONS & LOADS
! ================================================================
/SOLU
ANTYPE, 0                        ! static analysis

! Fix root
CSYS, 0
NSEL, S, LOC, Y, -1, 1
D, ALL, ALL, 0
ALLSEL, ALL

! Nodal loads from MATLAB (case D_fwd)
ALLSEL,ALL
/INPUT,htp_loads_D_fwd,inp
ALLSEL,ALL
```

7.1 Equilibrium check

In a linear static analysis the total reaction at the root must equal the total applied load. This is the single most useful check that the loads have actually reached the structure, and it is run after the solution:

```
/POST1
SET, LAST
ALLSEL,ALL
PRRSOL,FZ                        ! read the TOTAL VALUES line

! optional: total force and moment at the root
NSEL,S,LOC,Y,-1,1
FSUM                             ! moments are in N mm
ALLSEL,ALL
```

PRRSOL lists the reactions at the constrained nodes only, so the node numbers in that list are root node numbers and are unrelated to the loaded nodes. Only the TOTAL VALUES line at the foot of the list is of interest.

7.2 Results

Case	Applied load [N]	Root reaction FZ [N]	Root moment [N m]	Tip deflects
D_fwd	-9 142.9	+9 142.8 (confirmed)	-16 058	downward
C_fwd	-7 254	+7 254 (expected)	-12 741	downward
A_aft	+1 612	-1 611 (confirmed)	+2 831	upward

The applied load and the root reaction agree to within 1 N in both confirmed cases, which verifies the whole chain from the V-n diagram to the finite element model. Note that the model represents one half of the tail, so each case carries half of the total tail load of section 3.3.

> **Signs in ANSYS output.** PRRSOL reports the *reaction*, which opposes the applied load, so a download gives a positive reaction. FSUM reports the total of the *applied* loads and therefore carries the same sign as the tail load. FSUM moments are in N mm because the model is built in millimetres.

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8. Validity of the load files

The `.inp` files contain explicit node numbers. They are valid only for the exact mesh from which `htp_nodes.csv` was exported.

Change to the model	Load files still valid?	Action required
Thicknesses, stringer and flange dimensions	Yes	None. Sizing variables do not change the node numbering, so the files can be reused throughout a sizing optimisation.
Configuration (CONFIG = 1, 2 or 3)	No	Re-export the nodes and re-run step 4. The rib count and topology change the mesh.
Mesh density (ESIZE_V, rib element size)	No	Re-export the nodes and re-run step 4.
Macro-geometry (span, chords, dihedral)	No	Re-run steps 3 and 4 with the new geometry, then re-export the nodes.
Load case only	Yes	Change `case_to_export` in step 4 and re-run that script alone.

Units are newtons for forces and millimetres for positions, consistent with the N, mm, tonne, MPa system of the model. `FZ` is the global vertical direction.

9. Problems encountered and how they were resolved

Three defects were found and corrected during the work. They are recorded here because the first two will recur if the pipeline is rebuilt, and the third is a general trap in mixed shell and beam models.

9.1 Commas inside CSV text fields

Point descriptions written by step 1 originally contained commas, which split the affected rows into extra columns. MATLAB then silently dropped the six manoeuvre envelope points, and step 2 produced results for the gust cases only. All descriptions are now comma-free, and steps 2 and 3 check the expected row count (12 and 24 respectively) and stop with a clear message if it is not met.

9.2 Load lumped at the first loaded station

The first version of step 4 treated each station force as a point value of the polynomial. Because the first loaded station is rib 2, that station must carry all the load inboard of the first mid-bay cut, and the point-value formulation could not represent it. The first station received about -300 N instead of about -1 370 N, which placed the resultant too far outboard and overestimated the root bending moment by 17 per cent. Integrating the polynomial over tributary intervals, as described in section 6.1, corrected it to within rounding.

9.3 Beam-internal nodes in the load set

The original node export selected all nodes lying on the skin areas. Because the BEAM188 elements use cubic shape functions, that selection also captured their internal nodes. Of the 1 856 nodes exported, 804 belonged to beam elements. A force on such a node is not transmitted to the structure, so only 5 102 N of the 9 143 N applied reached the root, a loss of 44 per cent. Restricting the selection to the material 2 skin shell elements, as in section 5, reduced the list to 1 052 genuine skin nodes and brought the reaction into exact agreement with the applied load.

> **Diagnostic worth keeping.** If the root reaction does not match the applied load, list the applied forces with `FLIST,ALL` (redirected to a file with `/OUTPUT`), sum them, and compare the node set against `htp_nodal_forces.csv`. Any node whose attached element count or element type is unexpected should be excluded from the load set.

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10. Assumptions and limitations

The following must be stated in the project report, because they bound the validity of the results.

- **Balancing loads only.** The three cases are the tail loads required for pitch equilibrium. They do not include the manoeuvre and gust load increments of CS-25.331, 25.341 and 25.427, and no ultimate factor of safety of 1.5 has been applied. They are not a complete design load set.
- **Mixed data sources.** The flight loads are derived from ATR 42-600 data, while the tail geometry is that of Palumbo et al. (2026). The tail of the reference paper is larger than the ATR 42 tail, and the load has not been scaled for the difference in area. This is a known inconsistency of the present approach.
- **Assumed aerodynamic data.** The clean maximum lift coefficient, the negative maximum lift coefficient, the wing-body zero-lift pitching moment coefficient, the equality of V_C and V_{MO} and the gust velocities are assumptions, not confirmed ATR data. The sensitivity of the tail load to Cm_0 is reported by step 2.
- **No torsion.** The load at each rib is divided equally between that rib's nodes, so the chordwise position of the resultant is not controlled and the model is not loaded in torsion. This follows the agreed project scope but means the local chordwise stress distribution near the ribs is approximate, and a residual pitching moment appears in the `FSUM` output.

- **Quarter-chord sweep neglected.** The straight trailing edge places all of the taper on the leading edge, which sweeps the quarter-chord line by roughly 13 degrees. This would matter for a torsional analysis but does not affect the vertical shear force and bending moment.
- **Beam stresses excluded from the reported maximum.** The maximum von Mises stress extracted by the APDL script is taken from nodal averaging, which excludes beam and pipe elements. The stringers, spar flanges and rib flanges are therefore not represented in that figure.
- **Preliminary V-n diagram.** The envelope is built from public data and the assumptions above. It is not the certified flight envelope of the aircraft.

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11. File inventory

File	Type	Role
`vn_diagram.m`	MATLAB	Step 1. Writes vn_load_cases.csv and the V-n plot.
`tail_load.m`	MATLAB	Step 2. Reads vn_load_cases.csv, writes tail_load_cases.csv.
`htp_sfbm.m`	MATLAB	Step 3. Reads tail_load_cases.csv, writes htp_sfbm.csv.
`htp_nodal_forces.m`	MATLAB	Step 4. Reads htp_sfbm.csv and htp_nodes.csv, writes htp_loads_<case>.inp.
`vn_load_cases.csv`	data	12 V-n points: speed, load factor, weight.
`tail_load_cases.csv`	data	24 cases: 12 points at forward and aft CG, with L_H.
`htp_sfbm.csv`	data	w, V and M at the 14 rib stations, per case.
`htp_nodes.csv`	data	1 052 loaded skin nodes: node number and span position in mm.
`htp_nodal_forces.csv`	data	Solved nodal forces: node, span position, force in N.
`htp_loads_D_fwd.inp`	APDL	Load file, critical download case.
`htp_loads_C_fwd.inp`	APDL	Load file, cruise download case.
`htp_loads_A_aft.inp`	APDL	Load file, upload case.

The four MATLAB scripts must run in order, in the same folder, because each reads the CSV written by the previous one. The APDL node export block of section 5 belongs in the model script, between the final node merge and `/SOLU`.

To reproduce the load files for a new mesh: run the APDL model script to export `htp_nodes.csv`, copy it to the MATLAB folder, run `htp_nodal_forces.m` once per case, copy the resulting `.inp` files back to the ANSYS working directory, and verify each run with the root reaction check of section 7.1.